ADITYA MALVIYA

Senior Data Engineer | Azure Data Engineer | Azure Databricks, PySpark, Azure Data Factory, Delta Lake

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PROFESSIONAL SUMMARY

Senior Data Engineer with nearly 5 years of experience at Tata Consultancy Services (TCS), building production Azure data platforms for Rolls-Royce. Designs batch and real-time ETL/ELT pipelines processing ~27 GB/day with Azure Databricks, PySpark, Spark SQL, Azure Data Factory (ADF), ADLS Gen2, Delta Lake and Unity Catalog on a Medallion (Lakehouse) Architecture. Cut Spark job runtime by 73% and Azure costs by ~40%, and leads a 6-member team as Scrum Master. Microsoft Certified: Fabric Data Engineer Associate (DP-700).

TECHNICAL SKILLS

Languages: Python, SQL, T-SQL, PySpark, Spark SQL

Big Data & Processing: Apache Spark, Spark Structured Streaming, Apache Kafka, Spark Performance Tuning, Adaptive Query Execution (AQE), Data Skew Handling

Azure Data Platform: Azure Databricks, Azure Data Factory (ADF), Azure Data Lake Storage Gen2 (ADLS Gen2), Azure Event Hubs, Azure Blob Storage, Azure Key Vault, Azure DevOps, Azure API Management, Microsoft Fabric

Databricks & Lakehouse: Delta Lake, Unity Catalog, Databricks Workflows, Auto Loader, Delta MERGE, OPTIMIZE / Z-ORDER, Lakehouse Architecture, Medallion Architecture (Bronze, Silver, Gold)

Data Modeling & Warehousing: ETL, ELT, Data Pipelines, Data Warehousing, Dimensional Modeling, Star Schema, Slowly Changing Dimensions (SCD Type 2), Incremental Loads, Change Data Capture (CDC)

Data Governance & Quality: Data Quality Checks, Data Governance, Data Lineage, Role-Based Access Control (RBAC)

Databases & BI: Microsoft SQL Server, PostgreSQL, MySQL, Redis, DuckDB, Power BI

DevOps & Engineering: CI/CD (Azure DevOps Pipelines, YAML), Git, GitHub Actions, Docker, Pytest, SonarQube, FastAPI, REST APIs, dbt

Cloud (other): Google Cloud Platform (GCP), Databricks on GCP, Vertex AI, Google Cloud Functions

Methodologies & Leadership: Agile, Scrum Master, Sprint Planning, Code Reviews, Team Leadership, Mentoring, Stakeholder Management

Working Knowledge: Azure Synapse Analytics, Delta Live Tables (DLT), Apache Airflow

PROFESSIONAL EXPERIENCE

Senior Data Engineer | Tata Consultancy Services (TCS) | Mumbai, India | Dec 2021 – Present

Promoted from Data Engineer to Senior Data Engineer in Mar 2026. Clients: Rolls-Royce Digital (aerospace) and Avery Dennison (US Fortune 500).

Rolls-Royce Digital – Engine History Analysis, Azure Lakehouse Platform

- Design and own production Azure Data Factory (ADF) and Azure Databricks ETL/ELT pipelines in PySpark and Spark SQL on a Medallion Architecture over Delta Lake and ADLS Gen2, processing ~27 GB/day streaming and 5–7 GB/day batch data into star-schema models for Power BI.
- Built real-time ingestion with Azure Event Hubs, Apache Kafka and Spark Structured Streaming, plus incremental loads using Delta MERGE, SCD Type 2 and Auto Loader, orchestrated with Databricks Workflows.
- Cut a critical Spark job's runtime by 73% (52 to 14 minutes) by fixing join skew with key salting and Adaptive Query Execution (AQE), eliminating executor out-of-memory failures.
- Reduced Azure costs by ~40% through Databricks cluster right-sizing, Delta OPTIMIZE, Z-ORDER, partitioning and parameterized ADF pipelines.
- Implemented layer-wise data quality checks and Unity Catalog governance (RBAC, data lineage); secured credentials with Azure Key Vault-backed secret scopes and service principals.
- Promote ADF and Databricks code across dev, test and production with Azure DevOps CI/CD pipelines (YAML, Git branching).
- Lead a 6-member offshore team as Scrum Master: run stand-ups and sprint planning, manage Azure DevOps boards, mentor juniors, review pull requests and own delivery commitments to the client.

Rolls-Royce Digital – Assumptions Management, Data Services and APIs

- Built production FastAPI data services with RBAC, Azure API Management and Docker over SQL Server and Blob Storage; improved performance by ~30% with Redis caching and SQL/T-SQL query optimization.
- Raised code quality by resolving SonarQube findings and adding Pytest unit tests in Agile/Scrum sprints.

Avery Dennison (Smartrac RFID) – R&D Data and AI Platform

- Built Databricks pipelines on Google Cloud Platform (GCP) to ingest, cleanse and prepare RFID R&D datasets for training machine learning models behind an AI research tool now in production.
- Developed backend services with Vertex AI, Google Cloud Functions and a vector database for semantic search.

Python Developer Intern | BYJU'S (Gradeup) | Remote | Jul 2021 – Nov 2021

- Built Python scripts and REST API integrations for Gradeup, an exam-preparation platform acquired by BYJU'S.

FREELANCE WORK

Moriveda – D2C E-commerce Platform | moriveda.in **| May 2026 – Jun 2026**

- Built and launched a live e-commerce site with a Python/PHP backend and SQL database, integrating Razorpay payments and iThinkLogistics shipping APIs.
- Automated scheduled jobs for payment reconciliation, abandoned-cart recovery and review requests using cron.

PROJECTS

- **RepoReel – Reel-First Repository Search (Oct 2026 – Present):** Free nightly data pipeline (Python, DuckDB, fastembed, GitHub Actions) that finds English Instagram reels about GitHub repos, matches each to its repo via captions, cover OCR and GitHub search, and publishes sharded static data within free tiers; 411 reels, searchable by meaning. Live | GitHub
- **Corpus – Document AI and Data Q&A Workspace (Oct 2026 – Present):** Browser-native RAG app with hybrid BM25 + vector retrieval, intent routing and sentence-level citations, running Gemma 4 on WebGPU; turns questions about CSV/Parquet files into guarded DuckDB SQL and scores retrieval with hit@k and MRR. Live | GitHub
- **Orchestra – Pipeline Orchestrator and Run Doctor (Oct 2026 – Present):** Airflow-style scheduler on a simulated clock with cron, catch-up, retries with exponential backoff, pools, trigger rules, SLAs and idempotent backfills; diagnoses failures (root cause vs blast radius) and exports DAGs to Apache Airflow 2. Live | GitHub
- **FlowForge – Visual ETL Pipeline Builder (Oct 2026 – Present):** Drag-and-drop ETL designer that generates DuckDB SQL, Spark SQL, PySpark and dbt code with data-quality tests. Live | GitHub
- **Research: AQE vs Salting for Spark Skewed Joins (Aug 2026):** Reproducible benchmarks and cost models comparing Adaptive Query Execution and selective salting. GitHub
- **PipeScope – Spark Performance and Lineage Analyzer (Aug 2025 – Present):** Analyzes Spark event logs for tuning issues and traces SQL/dbt column-level lineage. Live | GitHub
- **DataLens – Data Profiling and Quality Tool (Jul 2024 – Present):** Profiles CSV, Parquet and JSON files with DuckDB-WASM and scores data completeness, validity and uniqueness. Live | GitHub
- **QueryRoom – SQL Practice Studio (Apr 2023 – Present):** In-browser SQLite workspace with 332 SQL interview questions, query execution and answer verification. Live | GitHub

CERTIFICATIONS

- Microsoft Certified: Fabric Data Engineer Associate (DP-700)
- Microsoft Certified: Azure Data Fundamentals (DP-900)

EDUCATION

Bachelor of Technology (B.Tech), Mechanical Engineering | Bhilai Institute of Technology, Durg | 2021
Grade: 71%